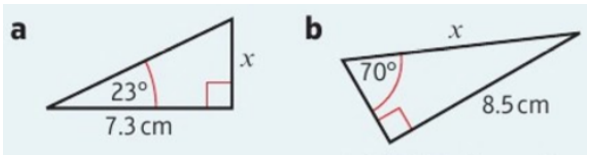
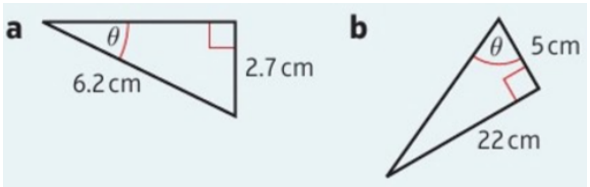
Chapter 9: Trigonometric ratios

Chapter 9 · Original slide 1 · Trigonometric ratios

Y1 Chapter 9
Trigonometric Ratios

Chapter 9 · Original slide 2 · Starters





1) Use trigonometry to find the lengths of the marked sides.

6) Find the sizes of the angles marked.

12) $f(x) = x^2 + 3x$. Sketch the graphs of:

m) $y = f(x)$

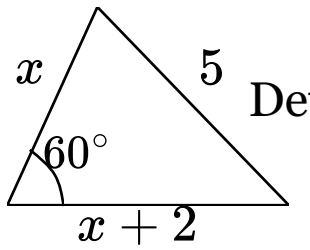
n) $y = f(x + 2)$

o) $y = f(x) - 3$

p) $y = f\left(\frac{1}{2}x\right)$

Chapter 9: Trigonometric ratios

Chapter 9 · Original slide 3 · Chapter Overview



Determine x.

There is technically no new content in this chapter since GCSE.

However, the problems might be more involved than at GCSE level.

1: Sine/Cosine Rule

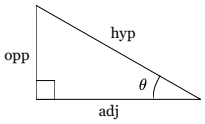
In ΔABC, $AB = 5, BC = 6$ and $\angle ABC = x$. Given that the area of $\angle ABC$ is 12cm2 and that AC is the longest side, find the value of x.

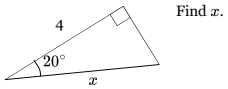
Sketch $y = \sin(2x)$ for $0 \leq x \leq 360^\circ$

2: Areas of Triangles

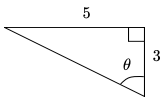
3: Graphs of Sine/Cosine/Tangent

Chapter 9 · Original slide 4 · RECAP: Right-Angled Trigonometry





Find x.



Find θ.

You are probably familiar with the formula: $\sin(\theta) = \frac{opp}{hyp}$

But what is the *conceptual* definition of $\sin$?

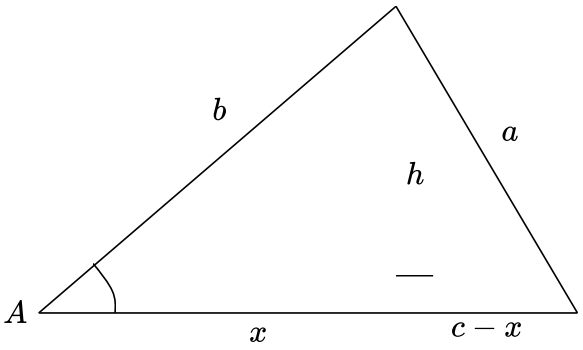
sin is a function which inputs an angle and gives the ratio between the opposite and hypotenuse.

Remember that a ratio just means the 'relative size' between quantities (in this case lengths). For this reason, sin/cos/tan are known as "trigonometric ratios".

reflections: You may have been taught "use tan^{-1} whenever you're finding an angle", and therefore write the second line directly. It is preferable to always write the first line, then see the problem as a 'changing the subject' one. We need to remove the tan on front of the θ, so apply tan^{-1} to each side of the equation to 'cancel out' the tan on the LHS.

Tip: You can swap the thing you're dividing by and the result. e.g. $\frac{8}{2} = 4 \rightarrow \frac{8}{4} = 2$.

Chapter 9 · Original slide 5 · Proof of Cosine Rule



We want to use Pythagoras, so split c into two so that we get two right-angled triangles.

By Pythagoras: $h^2 + x^2 = b^2$ and
$h^2 + (c - x)^2 = a^2$

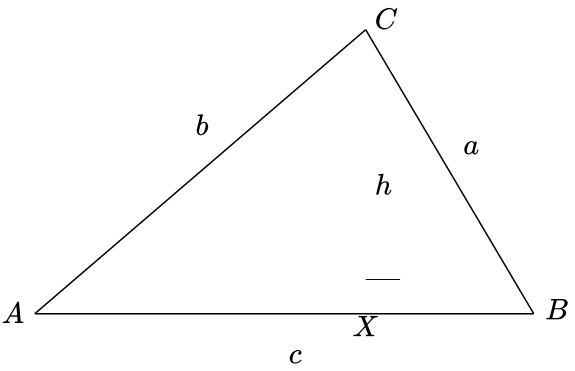
Subtracting to eliminate h:

$x^2 - (c - x)^2 = b^2 - a^2$
$a^2 = b^2 + c^2 - 2cx$

But $x = b\cos A$ $\therefore$

$a^2 = b^2 + c^2 - 2bc\cos A$

Chapter 9 · Original slide 6 · Proof of Sine Rule



The idea is that we can use the common length of ΔACX and $\angle XBC$, i.e. h, to connect the two triangles, and therefore connect their angles/length.

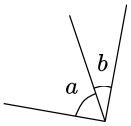
Using basic trigonometry:

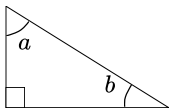
$h = a\sin B$ and $h = b\sin A$

$\therefore a\sin B = b\sin A$
$\therefore \frac{a}{\sin A} = \frac{b}{\sin B}$

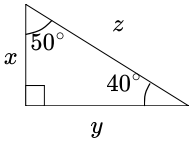
Chapter 9: Trigonometric ratios

Chapter 9 · Original slide 7 · Just for your interest…





Why does “cosine” contains the word “sine”?



Complementary Angles

add to 90°

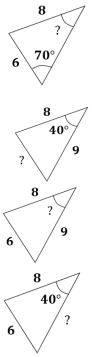
Therefore these angles are complementary.

i.e. The cosine of an angle is the sine of the complementary angle.

Hence cosine = COMPLEMENTARY SINE

$$\therefore \mathbf{cos}\,(\mathbf{50}) = \mathbf{sin}\,(\mathbf{40})$$

Chapter 9 · Original slide 8 · OVERVIEW: Finding missing sides and angles

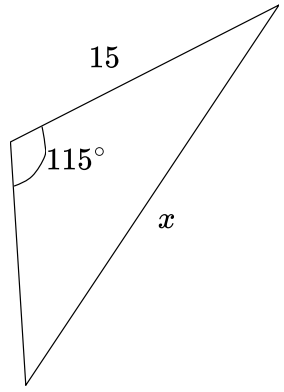


When triangles are not right-angled, we can no longer use simple trigonometric ratios, and must use the cosine and sine rules.

You have	You want	Use
#1: Two angle-side opposite pairs	Missing angle or side in one pair	
#2 Two sides known and a missing side opposite a known angle	Remaining side	
#3 All three sides	An angle	
	Remaining side	

Chapter 9: Trigonometric ratios

Chapter 9 · Original slide 9 · Cosine Rule



We use the cosine rule whenever we have three sides (and an angle) involved.

b

Cosine Rule:

$a^2 = b^2 + c^2 - 2bc\cos A$

Proof at end of PowerPoint.

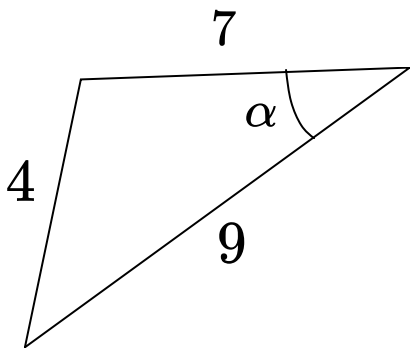
a A

The only angle in formula is A, so label angle in diagram A, label opposite side a, and so on (b and c can go either way).

c

12

Chapter 9 · Original slide 10 · Dealing with Missing Angles



You have	You want	Use
All three sides	An angle	Cosine rule

$a^2 = b^2 + c^2 - 2bc\cos A$

Tip: The brackets are not needed, but students who forget about BIDMAS see $4^2 = \left(7^2 + 9^2 - 2 \times 7 \times 9\right)\cos\alpha$ and hence incorrectly simplify to $16 = 4\cos\alpha$

Label sides then substitute into formula.

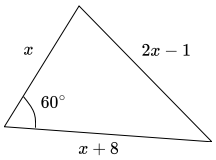
Simplify each bit of formula.

Rearrange "swapping" thing you're subtracting and result

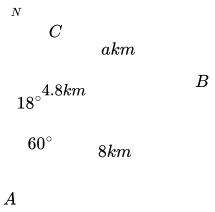
Textbook Note: The textbook presents the rearrangement of the cosine rule: $\cos A = \frac{b^2+c^2-a^2}{2bc}$ to find missing angles. It is not advisable to memorise this

Chapter 9: Trigonometric ratios

Chapter 9 · Original slide 11 · Harder Ones

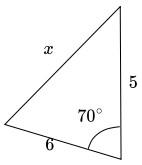


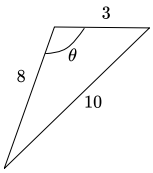
Determine the value of x.

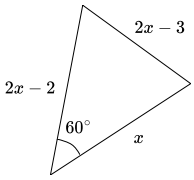


[From textbook] Coastguard station B is 8 km, on a bearing of 060°, from coastguard station A. A ship C is 4.8 km on a bearing of 018°, away from A. Calculate how far C is from B.

Chapter 9 · Original slide 12 · Test Your Understanding





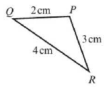


1 3 2

Note: You will get an obtuse angle whenever you inverse cos of a negative value.

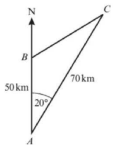
Chapter 9 · Original slide 13 · Exercise 9A – Y1 pages 177-179

8 Show that $\cos P = -\frac{1}{4}$.



11 The lengths of the sides of a triangle are in the ratio $2 : 3 : 4$. Calculate the size of the largest angle.

Chapter 9 · Original slide 14 · Exercise 9A – Y1 pages 177-179



17 The diagram shows three cargo ships, A, B and C, which are in the same horizontal plane. Ship B is 50 km due north of ship A and ship C is 70 km from ship A. The bearing of C from A is $020°$.
a Calculate the distance between ships B and C, in kilometres to 3 s.f. (3 marks)
b Calculate the bearing of ship C from ship B. (4 marks)

Chapter 9: Trigonometric ratios

Chapter 9 · Original slide 15 · Extension

1

[STEP I 2009 Q4i]

The sides of a triangle have lengths $p-q$, p and $p+q$, where $p>q>0$. The largest and smallest angles of the triangle are α and β respectively. Show by means of the cosine rule that

$4\left(1-\cos\alpha\right)\left(1-\cos\beta\right)=\cos\alpha+\cos\beta$

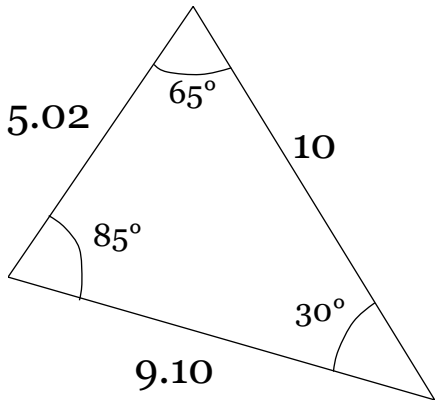
Note that the longest side of a triangle is opposite the largest angle, and the shortest opposite the smallest angle. Thus:

α $p-q$

β p

$p+q$

Chapter 9 · Original slide 16 · The Sine Rule



C B a

! Sine Rule:

$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}$

c b A

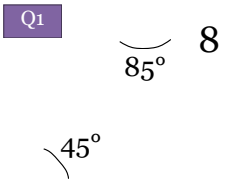
For this triangle, try calculating each side divided by the sin of its opposite angle. What do you notice in all three cases?

You have	You want	Use
#1: Two angle-side opposite pairs	Missing angle or side in one pair	Sine rule

Chapter 9: Trigonometric ratios

Chapter 9 · Original slide 17 · Examples

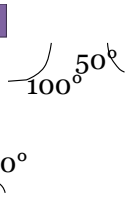
Q1



$\frac{x}{\sin 85} = \frac{8}{\sin 45}$

$x = \frac{8 \sin 85}{\sin 45} = 11.27$

Q2



8

$\frac{x}{\sin 100} = \frac{8}{\sin 30}$

$x = \frac{8 \sin 100}{\sin 30} = 15.76$

You have	You want	Use
#1: Two angle-side opposite pairs	Missing angle or side in one pair	Sine rule

Chapter 9 · Original slide 18 · Examples

Q3



$\frac{\sin \theta}{8} = \frac{\sin 126^\circ}{10}$

$\sin \theta = \frac{8 \sin 126}{10}$

$\theta = \sin^{-1}\left(\frac{8 \sin 126}{10}\right)$

$= 40.33^\circ$

Q4



When you have a missing angle, it's better to reciprocate to get:

$\frac{\sin A}{a} = \frac{\sin B}{b}$

i.e. in general put the missing value in the numerator.

$\frac{\sin \theta}{5} = \frac{\sin 85}{6}$

$\sin \theta = \frac{5 \sin 85}{6}$

$\theta = \sin^{-1}\left(\frac{5 \sin 85}{6}\right)$

$= 56.11^\circ$

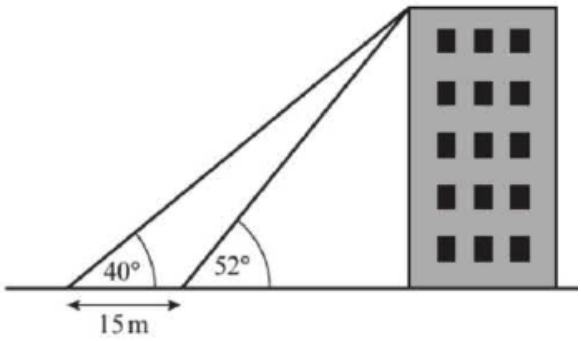
Chapter 9 · Original slide 19 · Exercise 9B – Y1 pages 181-183

8 Town B is 6 km, on a bearing of 020°, from town A. Town C is located on a bearing of 055° from town A and on a bearing of 120° from town B.
Work out the distance of town C from:
a town A $btownB$

Chapter 9 · Original slide 20 · Exercise 9B – Y1 pages 181-183

11 In $\triangle ABC$, $x = 4(\sqrt{2} - 1)$
$AB = x$ cm, $BC = (4 - x)$ cm, $\angle BAC = y$ and $\angle BCA = 30^\circ$.
Given that $\sin y = \frac{1}{\sqrt{2}}$, show that

Chapter 9 · Original slide 21 · Exercise 9B – Y1 pages 181-183

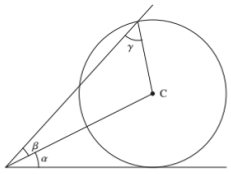


12 A surveyor wants to determine the height of a building. She measures the angle of elevation of the top of the building at two points 15 m apart on the ground.

a Use this information to determine the height of the building. (4 marks)

b State one assumption made by the surveyor in using this mathematical model. (1 mark)

Chapter 9 · Original slide 22 · Extension





If we draw a vertical line down from **C**, we have two triangles with a common length. This common lengths allows us to relate the two triangles. Let the radius be 1.

1

[MAT 2011 1E]

The circle in the diagram has centre C. Three angles α, β, γ are also indicated.

The angles α, β, γ are related by the equation:

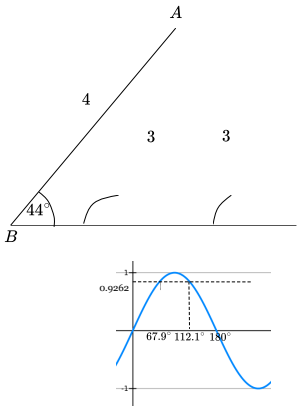
A) $\cos\alpha = \sin(\beta + \gamma)$

B) $\sin\beta = \sin\alpha \sin\gamma$

C) $\sin\beta\,(1 - \cos\alpha) = \sin\gamma$

D) $\sin(\alpha + \beta) = \cos\gamma \sin\alpha$

Chapter 9 · Original slide 23 · The ‘Ambiguous Case’



Suppose you are told that $AB = 4$, $AC = 3$ and $\angle ABC = 44°$. What are the possible values of $\angle ACB$?

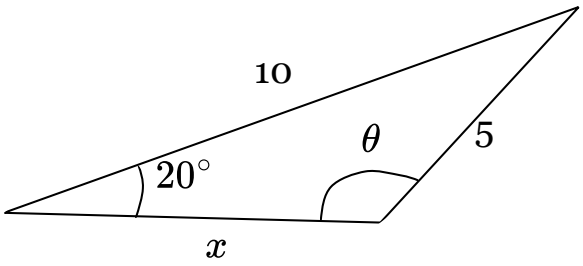
Your calculator will give the acute angle of $67.9°$ (i.e. C_2). But if we look at a graph of sin, we can see there’s actually a second value for $\sin^{-1}(0.9262)$, corresponding to angle C_1.

! The sine rule produces two possible solutions for a missing angle:

$\sin\theta = \sin(180° - \theta)$

Whether we use the acute or obtuse angle depends on context.

Chapter 9 · Original slide 24 · Test Your Understanding



Given that the angle θ is obtuse, determine θ and hence determine the length of x.

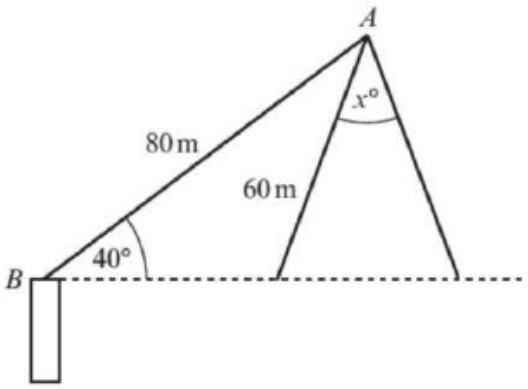
Chapter 9 · Original slide 25 · Exercise 9C – Y1 pages 184-185

4 Triangle ABC is such that $AB = 4$ cm, $BC = 6$ cm and $\angle ACB = 36°$. Show that one of the possible values of $\angle ABC$ is $25.8°$ (to 3 s.f.). Using this value, calculate the length of AC.

Chapter 9 · Original slide 26 · Exercise 9C – Y1 pages 184-185

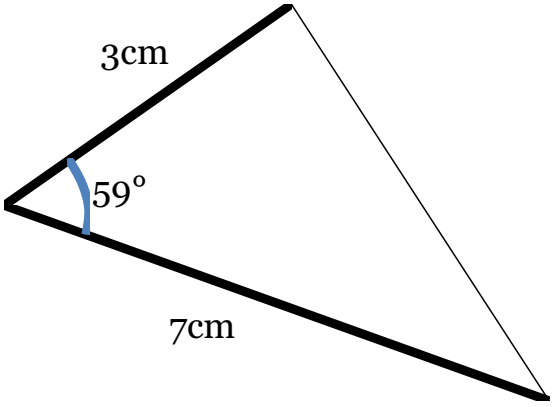
5 Two triangles ABC are such that $AB = 4.5$ cm, $BC = 6.8$ cm and $\angle ACB = 30°$. Work out the value of the largest angle in each of the triangles.

Chapter 9 · Original slide 27 · Exercise 9C – Y1 pages 184-185



6 a A crane arm AB of length 80 m is anchored at point B at an angle of 40° to the horizontal. A wrecking ball is suspended on a cable of length 60 m from A. Find the angle x through which the wrecking ball rotates as it passes the two points level with the base of the crane arm at B. (6 marks)
b Write down one modelling assumption you have made. (1 mark)

Chapter 9 · Original slide 28 · Area of Non Right-Angled Triangles

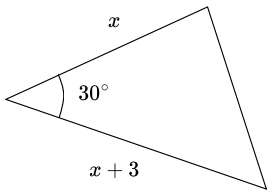


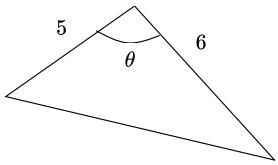
Area = $\frac{1}{2}ab\sin(C)$

where C is the angle between two sides a and b.

Tip: You shouldn't have to label sides/angles before using the formula. Just remember that the angle is between the two sides.

Chapter 9 · Original slide 29 · Test Your Understanding





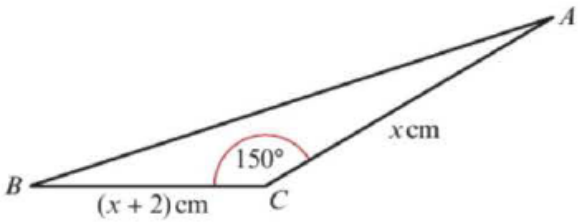
The area of this triangle is also 10.

If θ is obtuse, determine θ.

The area of this triangle is 10.

Determine x.

Chapter 9 · Original slide 30 · Exercise 9D – Y1 pages 186-187



4 In triangle ABC, $BC = (x+2)$ cm, $AC = x$ cm and $\angle BCA = 150^\circ$. Given that the area of the triangle is 5 cm^2, work out the value of x, giving your answer to 3 significant figures.

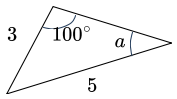
Chapter 9 · Original slide 31 · Exercise 9D – Y1 pages 186-187

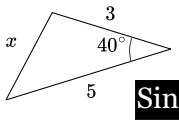
5 In $\triangle PQR$, $PQ = (x + 2)$ cm, $PR = (5 - x)$ cm and $\angle QPR = 30^\circ$. The area of the triangle is A cm^2.
a Show that $A = \frac{1}{4}(10 + 3x - x^2)$.
b Use the method of completing the square, or otherwise, to find the maximum value of A, and give the corresponding value of x.

Chapter 9 · Original slide 32 · Exercise 9D – Y1 pages 186-187

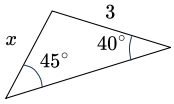
6 In $\triangle ABC$, $AB = x$ cm, $AC = (5 + x)$ cm and $\angle BAC = 150^\circ$. Given that the area of the triangle is $3\frac{3}{4}$ cm^2
a Show that x satisfies the equation $x^2 + 5x - 15 = 0$.
b Calculate the value of x, giving your answer to 3 significant figures.

Chapter 9 · Original slide 33 · Sin or cosine rule?

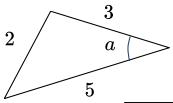




Sine



Sine Cosine



Sine

Recall that whenever we have two “side-angle pairs” involved, use sine rule. If there’s 3 sides involved, we can use cosine rule. Sine rule is generally easier to use than cosine rule.

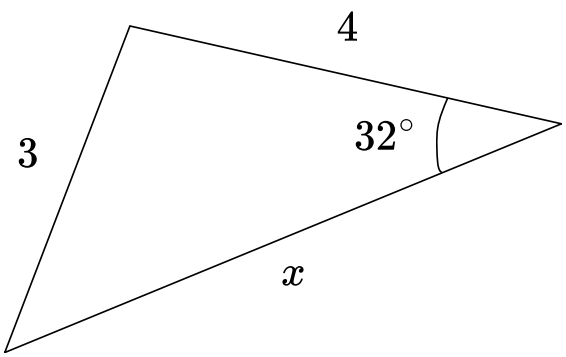
Sine

Cosine

Cosine

Cosine

Chapter 9 · Original slide 34 · Using sine rule twice



You have	You want	Use
#4 Two sides known and a missing side not opposite known angle	Remaining side	Sine rule twice

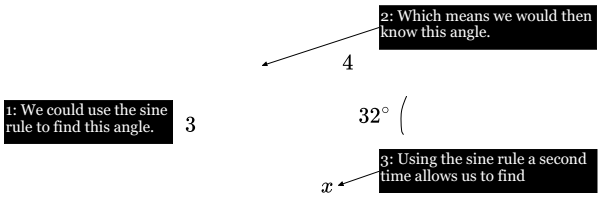
Given there is just one angle involved, you might attempt to use the cosine rule:

This is a quadratic equation!

It’s possible to solve this using the quadratic formula (using $\mathbf{a=1, b=-8\cos 32, c=7}$). However, this is a bit fiddly and not the primary method expected in the exam...

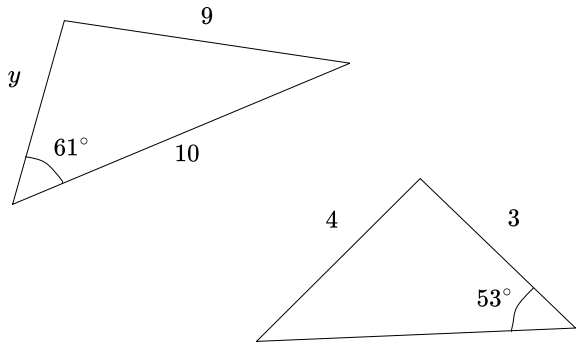
Chapter 9: Trigonometric ratios

Chapter 9 · Original slide 35 · Using sine rule twice

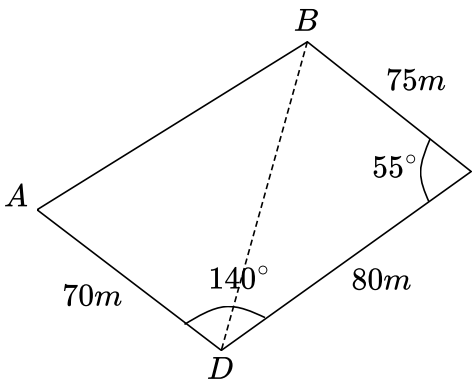


You have	You want	Use
#4 Two sides known and a missing side not opposite known angle	Remaining side	Sine rule twice

Chapter 9 · Original slide 36 · Test Your Understanding



Chapter 9 · Original slide 37 · Problem Solving With Sine/Cosine Rule



[From Textbook] The diagram shows the locations of four mobile phone masts in a field, $BC = 75m$.

$CD = 80m$, angle $BCD = 55°$ and angle $ADC = 140°$.

In order that the masts do not interfere with each other, they must be at least 70m apart.

Given that A is the minimum distance from D, find:

a) The distance A is from B

b) The angle BAD

c) The area enclosed by the four masts.

Chapter 9 · Original slide 38 · Exercise 9E – Y1 page 189-191

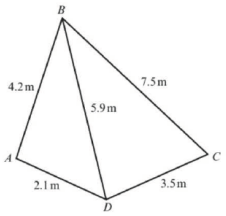
12 The perimeter of $\triangle ABC = 15$ cm. Given that $AB = 7$ cm and $\angle BAC = 60°$, find the lengths of AC and BC and the area of the triangle.

Chapter 9 · Original slide 39 · Exercise 9E – Y1 page 189-191

13 In the triangle ABC, $AB = 14$ cm, $BC = 12$ cm and $CA = 15$ cm.
a Find the size of angle C, giving your answer to 3 s.f.
b Find the area of triangle ABC, giving your answer in cm^2 to 3 s.f.

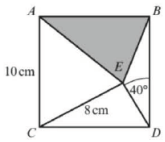
Chapter 9 · Original slide 40 · Exercise 9E – Y1 page 189-191

14 A flower bed is in the shape of a quadrilateral as shown in the diagram.
a Find the sizes of angles DAB and BCD. (4 ma...
b Find the total area of the flowe...
(3 ma...
c Find the length of the diagonal ...



Chapter 9 · Original slide 41 · Exercise 9E – Y1 page 189-191

15 $ABCD$ is a square. Angle CED is obtuse.
Find the area of the shaded triangle. (7 marks)



Chapter 9 · Original slide 42 · Extension Problem 1

[AEA 2009 Q5a] The sides of the triangle ABC have lengths $BC = a$, $AC = b$ and $AB = c$, where $a < b < c$. The sizes of the angles A, B and C form an arithmetic sequence.

(i) Show that the area of triangle ABC is $ac\frac{\sqrt{3}}{4}$.

Given that $a = 2$ and $\sin A = \frac{\sqrt{15}}{5}$, find

(ii) the value of b,

(iii) the value of c.

Chapter 9 · Original slide 43 · Extension Problem 2

[STEP I 2006 Q8] Note that the volume of a tetrahedron is equal to $\frac{1}{3} \times areaofbase \times height$

The points O, A, B, C have coordinates $(0,0,0), (a,0,0), (0,b,0)$ and $(0,0,c)$ respectively, where a, b, c are positive.

(i) Find, in terms of a, b, c the volume of the tetrahedron $OABC$.

(ii) Let angle $ACB = \theta$. Show that $\cos\theta = \frac{c^2}{\sqrt{(a^2+c^2)(b^2+c^2)}}$ and find, in terms of a, b and c, the area of triangle ABC.
Hence show that d, the perpendicular distance of the origin from the triangle ABC, satisfies $\frac{1}{d^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$

This is FM content, but see a few lines below.

Chapter 9 · Original slide 44 · Sin Graph

90 180 270 360

What does it look like?

Key Features:

• Repeats every $360°$.

• Range: $-1 < \sin(x) < 1$

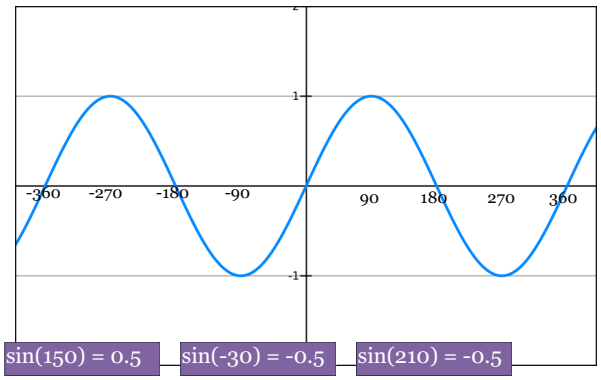
-360

-270

-180

-90

Chapter 9 · Original slide 45 · Sin Graph



What do the following graphs look like?

Suppose we know that sin(30) = 0.5. By thinking about symmetry in the graph, how could we work out:

Chapter 9 · Original slide 46 · Cos Graph

90 180 270 360

What do the following graphs look like?

Key Features:

- Repeats every 360°.
- Range: $-1 < \sin(x) < 1$

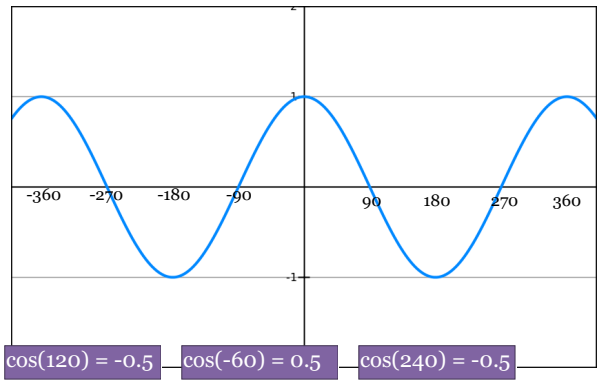
-360

-270

-180

-90

Chapter 9 · Original slide 47 · Cos Graph



What does it look like?

Suppose we know that cos(60) = 0.5. By thinking about symmetry in the graph, how could we work out:

Chapter 9 · Original slide 48 · Tan Graph

90 180 270 360

What does it look like?

-360

-270

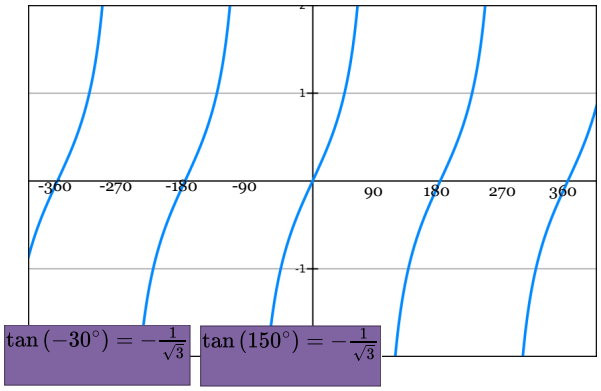
-180

-90

Key Features:

- Repeats every $\mathbf{180}^\circ$.
- Roots: $0, 180^\circ, -180^\circ, \ldots$
- Range: $\tan(x) \in \mathbb{R}$ (i.e. no min/max value!)
- Asymptotes: $x = \pm 90^\circ, \pm 270^\circ, \ldots$

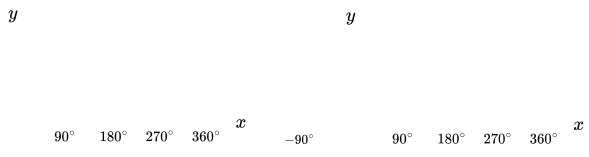
Chapter 9 · Original slide 49 · Tan Graph



What does it look like?

Suppose we know that $\tan(30°) = \frac{1}{\sqrt{3}}$. By thinking about symmetry in the graph, how could we work out:

Chapter 9 · Original slide 50 · Transforming Trigonometric Graphs



Sketch $y = 4\sin x$, $0 \leq x \leq 360°$

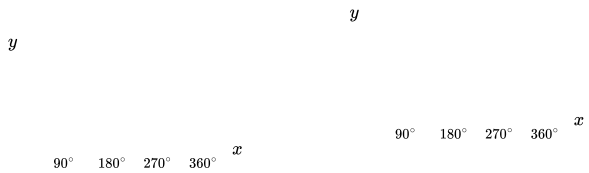
There is no new theory here: just use your knowledge of transforming graphs, i.e. whether the transformation occurs 'inside' the function (i.e. input modified) or 'outside' the function (i.e. output modified).

Sketch $y = \cos(x + 45°)$, $0 \leq x \leq 360°$

Chapter 9: Trigonometric ratios

Chapter 9 · Original slide 51 · Transforming Trigonometric Graphs

Sketch $y = -\tan x, 0 \leq x \leq 360^\circ$

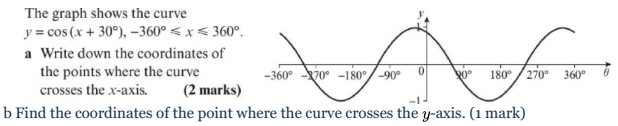


Sketch $y = \sin\left(\frac{x}{2}\right), 0 \leq x \leq 360^\circ$

Chapter 9 · Original slide 52 · Exercise 9F – Y1 Page 194

1 Sketch the graph of $y = \cos\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.
2 Sketch the graph of $y = \tan\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.
3 Sketch the graph of $y = \sin\theta$ in the interval $-90^\circ \leq \theta \leq 270^\circ$.

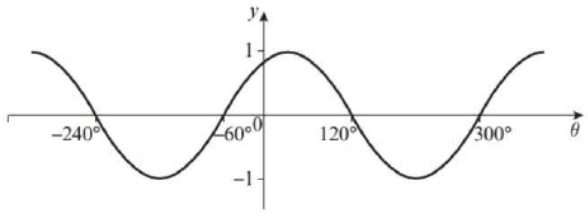
Chapter 9 · Original slide 53 · Exercise 9G – Y1 Pages 197-198



7 The graph shows the curve $y = \cos(x + 30^\circ), -360^\circ \leq x \leq 360^\circ$.
a Write down the coordinates of the points where the curve crosses the x-axis. (2 marks)

b $\cos 30^\circ = \dfrac{\sqrt{3}}{2}$; $\left(0, \dfrac{\sqrt{3}}{2}\right)$

Chapter 9 · Original slide 54 · Exercise 9G – Y1 Pages 197-198

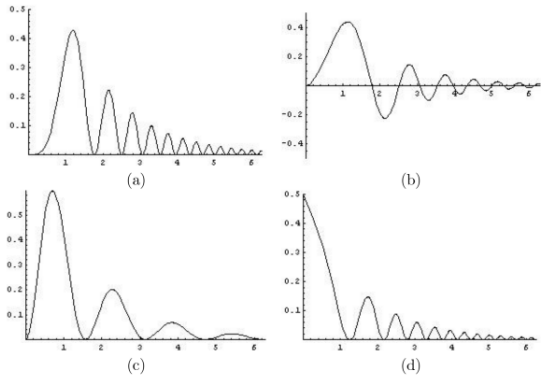


8 The graph shows the curve with equation $y = \sin(x + k), -360^\circ \leq x \leq 360^\circ$, where k is a constant.
a Find one possible value for k. (2 marks)
b Is there more than one possible answer to part a? Give a reason for your answer. (2 marks)

Chapter 9 · Original slide 55 · Exercise 9G – Y1 Pages 197-198

9 The variation in the depth of water in a rock pool can be modelled using the function $y = \sin(30t)^\circ$, where t is the time in hours and $0 \leq t \leq 6$.
a Sketch the function for the given interval.
b If $t = 0$ represents midday, during what times will the rock pool be at least half full?

Chapter 9 · Original slide 56 · Extension



1

[MAT 2013 1B] The graph of $y = \sin x$ is reflected first in the line $x = \pi$ and then in the line $y = 2$. The resulting graph has equation:

A) $y = \cos x$

B) $y = 2 + \sin x$

C) $y = 4 + \sin x$

D) $y = 2 - \cos x$

3

[MAT 2007 1G] On which of the axes is a sketch of the graph

$y = 2^{-x}\sin^2\left(x^2\right)$

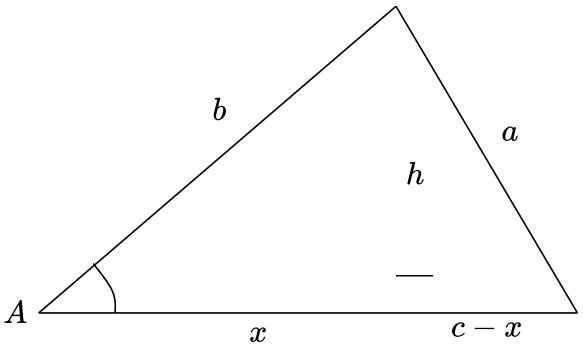
2

[MAT 2011 1D] What fraction of the interval $0 \leq x \leq 360^\circ$ is one (or both) of the inequalities:

$\sin x \geq \frac{1}{2}, \qquad \sin 2x \geq \frac{1}{2}$

true?

Chapter 9 · Original slide 57 · APPENDIX :: Proof of Cosine Rule



We want to use Pythagoras, so split c into two so that we get two right-angled triangles.

By Pythagoras: $h^2 + x^2 = b^2$ and
$h^2 + (c - x)^2 = a^2$

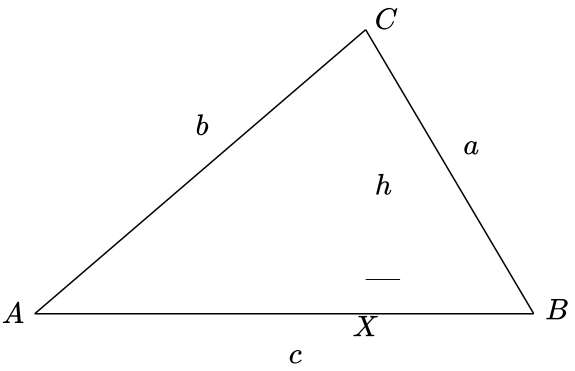
Subtracting to eliminate h:

$x^2 - (c - x)^2 = b^2 - a^2$
$a^2 = b^2 + c^2 - 2cx$

But $x = b\cos A$ $\therefore$

$a^2 = b^2 + c^2 - 2bc\cos A$

Chapter 9 · Original slide 58 · APPENDIX :: Proof of Sine Rule



The idea is that we can use the common length of ΔACX and $\angle XBC$, i.e. h, to connect the two triangles, and therefore connect their angles/length.

Using basic trigonometry:

$h = a\sin B$ and $h = b\sin A$

$\therefore a\sin B = b\sin A$
$\therefore \frac{a}{\sin A} = \frac{b}{\sin B}$